

How is generative AI gonna cause us to go into a water shortage when water is renewable? what does AI even need clean water for?

Posted by u/R3dcherries • 0 points • 6 comments

i keep seeing people talking about this but i didn't understand the "how" and "why" behind it all so i went to go google it but im still too stupid to get it.

Comments

u/Delehal • 0 points • 2026-03-15 19:09 UTC

AI needs a lot of computing power, such as massive new data centers that are being built around the world. Data centers need cooling. A lot of them use water for evaporative cooling.

In some areas, local utilities are being overwhelmed by spikes in water demand. Especially if local water sources run dry and they need to either ration what they have or bring in water from outside sources. That can drive up prices for everyone in the region. Some towns have felt screwed over by this.

At a global scale, though, you're right. Water is renewable. The water isn't destroyed. It goes into a cycle. We eventually get it back. Just keep in mind that takes time and there is only so much fresh water ready to use at any specific time in any given area.

With all that said, AI water usage has become sort of a weird pop culture thing. A lot of people have heard that it's an issue, so they carp on it, but data centers account for well under 1% of total water usage. Other industries are using way more water than AI.

u/Rabbitical • 2 points • 2026-03-15 17:54 UTC

Used cooling water only has two endings, either it evaporates back into the atmosphere, or it has to be cleaned at a treatment facility before being returned to the environment or used for anything else, which costs money and power. Water that's evaporated back into the atmosphere is yes, technically "renewable" but that doesn't mean it goes back to where it came from. Clean drinking water comes from very specific places, and those places only receive so much rain whether we "recycle" water or not. Since most of the planet is covered in ocean, that's where most of the rain ends up, and so is completely unusable to us. So, returning water to the environment doesn't magically refill our reservoirs.

u/PastelVice • 1 points • 2026-03-15 17:50 UTC

its not that AI drinks water or something lol. the data centers that run AI get insanely hot, and they use a lot of water to cool the machines. that water often has to be very clean so it doesnt damage the systems. if tons of new AI centers keep being built, the total water use can get pretty big

u/Commercial_Raise3378 • 0 points • 2026-03-15 17:46 UTC

Water is only renewable to a point.

u/chubbygrannychaser • 1 points • 2026-03-15 17:43 UTC

Water is in a global cycle. Nobody gets to drink fresh water tied up in glaciers or falling as rain over the ocean. People in Texas can't drink water falling in monsoons over India.

Even if the water does cycle back to the same area, we require the use of energy, infrastructure, chemicals, and other costs before we can actually consume that water. Those costs add up over time, making water expensive.

AI runs on large computer farms that require cheap efficient cooling. They use water to do that. Salt water causes corrosion. Unfiltered water allows debris to accumulate and cause clogs. Someone has to clean up and filter water before it is useful. Using the water this way takes it out of circulation for other things we might use it for. It might be kept as thousands of gallons circulating - with periodic top offs to replace losses. It may be too warm, it may pick up trace chemicals, or other reasons it cannot just be returned to rivers or streams.

u/TheKaritha • 1 points • 2026-03-15 17:42 UTC

AI doesn't use water directly, mostly it uses it for cooling of just like how you do cool off by sweating. It's getting used to cool off some data centers and electricity centers.

So is water renewable, yes. Should we renew it, no. Only way to get that used water again is by collection steam which might be polluted by chemicals from thermal stations. Even if we somehow get unpolluted water, cost of re-mineralizing water and cool it off is not cheaper than directly taking it from a river.

u/lostfornames • 2 points • 2026-03-15 17:38 UTC

AI uses clean water for cooling. It cant use salt water, its corrosive. The used water is now dirty and needs to be cleaned. This makes water more expensive and can be a strain on local water sources.